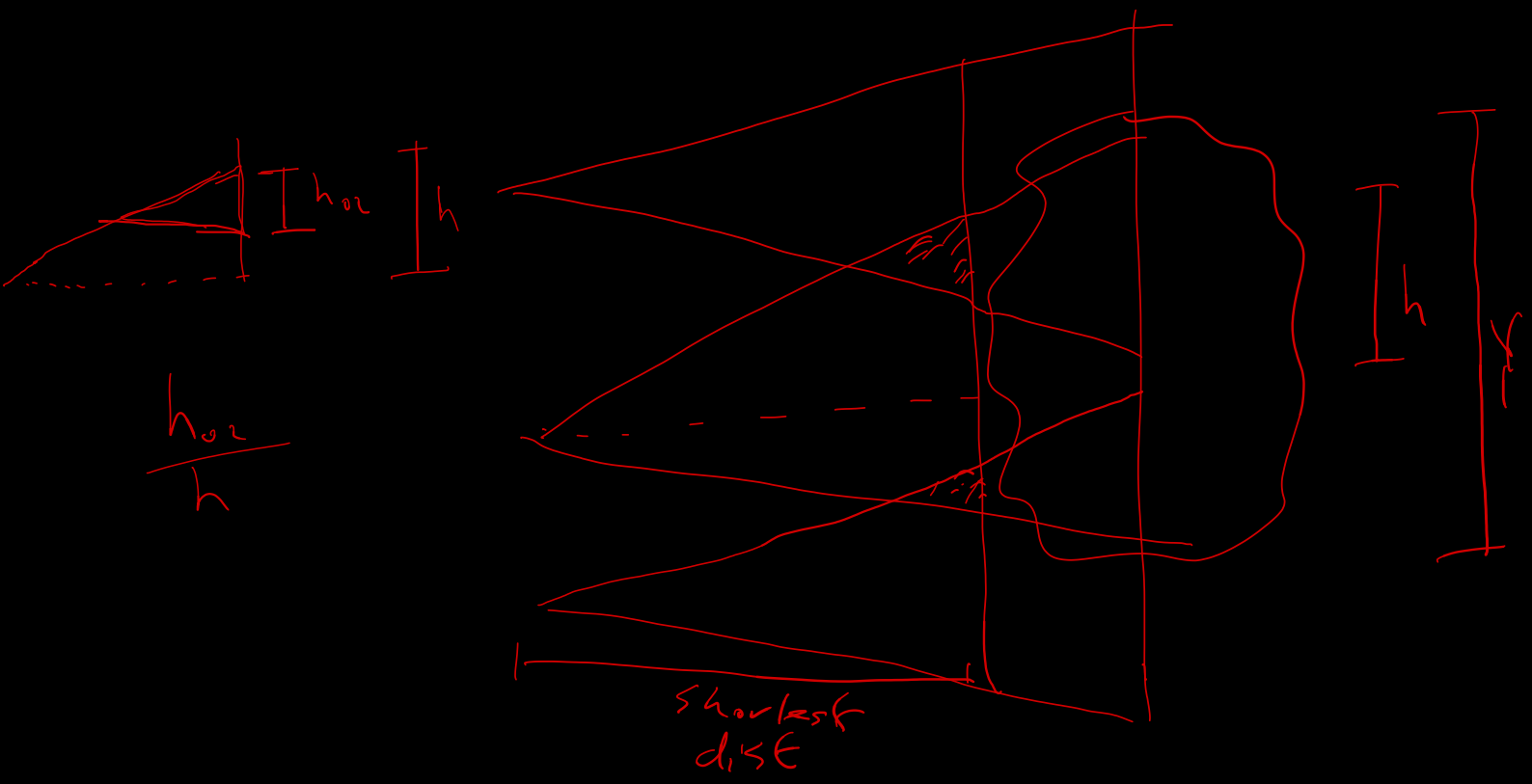


- Target a specific overlap.
- Have a given starting point on the vertical axes

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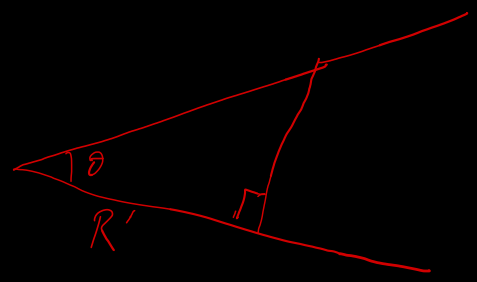
- Consider a target object with height $Y$
- Let the shortest distance be $R'$
- If the camera is placed at the lowest point on the vertical axis and its FOV is such that the lowest end of the object is seen and nothing below it, then the length of the plane at $R'$ that is seen by the camera is $h$




$T = \frac{0}{A}$

$h = R' \times \tan\theta$

$$\frac{h_{ov}}{h} = \text{Overlap} = Y'$$

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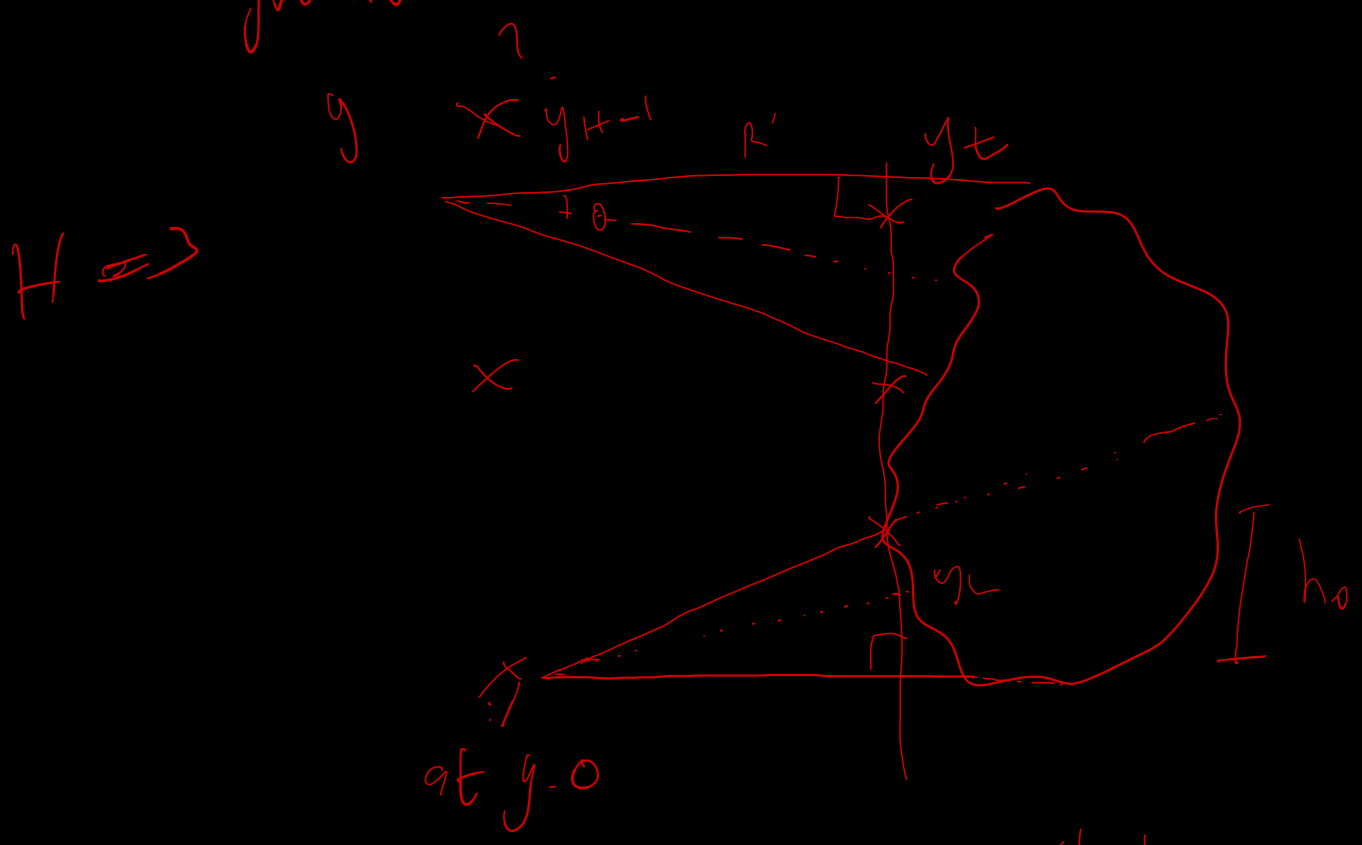
If $Y' = 0.7$ then $H = \frac{Y'}{0.7h}$

$\Rightarrow H = \text{ceil}\left(\frac{Y'}{Y'h}\right)$

N/B

- The plane at centroid-y perpendicular to the vertical axis is not a symmetrical plane w.r.t the height of the target.

* The camera always starts on the ground.




$y_t = Y - y_L$

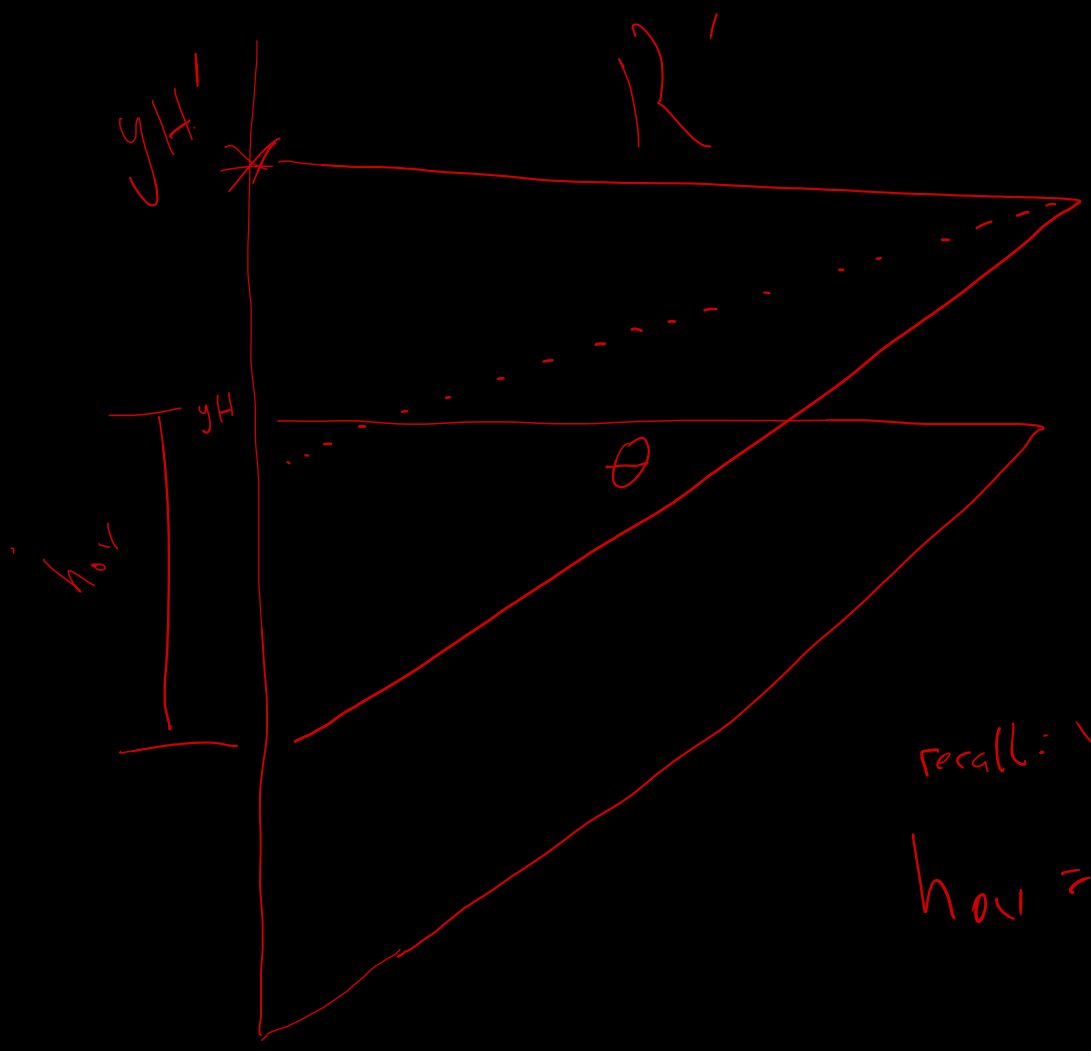
$Y' \times h_0$

## Challenge

- The topmost tip will not always be reconstructed.
- Solution: We offset it.

How? We increase Y.

By how much?




$yH' = ?$

recall: $Y' = \frac{h_{ov}}{h}$

$h_{ov} = Y' \times h_o$

Consider $yH'$ & $yt$.

To get $yH' = yH - h_{ov} + h$

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